D. Y. PATIL COLLEGE OF ENGINEERING AND TECHNOLOGY, KOLHAPUR

(An Autonomous Institute)

DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING

A

PROJECT

REPORT

ON

"FULL-WAVE RECTIFIER WITH 7805 REGULATED POWER SUPPLY ON PCB"

SUBMITTED BY:

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1. Objective

To design and fabricate a Full-Wave Rectifier circuit with a 7805 voltage regulator for obtaining a stable +5 V DC output from an AC input source.

2. Introduction

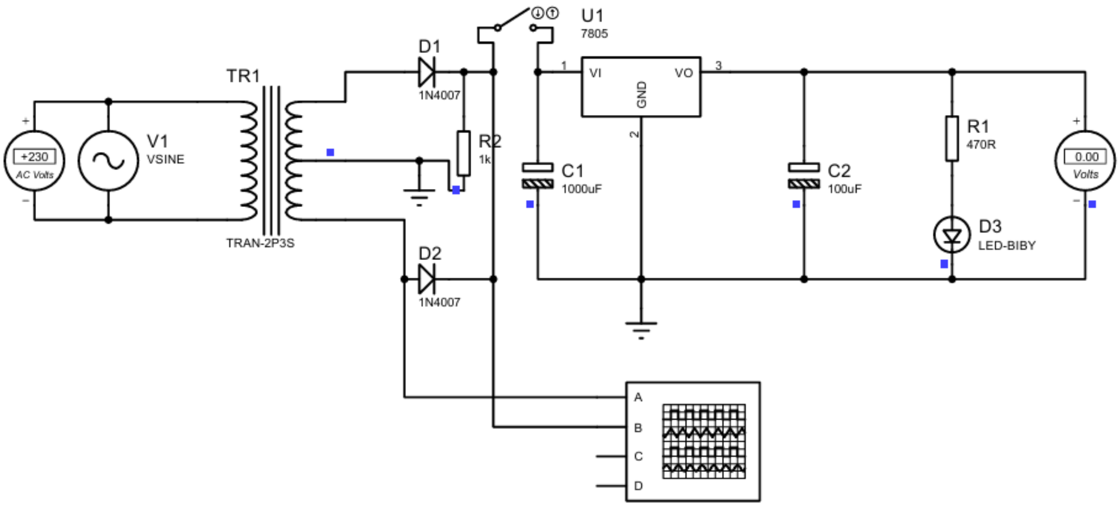
A rectifier is an electronic circuit used to convert Alternating Current (AC) into Direct Current (DC). In this project, a centre-tapped full-wave rectifier is implemented using two diodes. The pulsating DC obtained from the rectifier is filtered using capacitors and regulated using the 7805 voltage regulator IC to produce a constant +5 V DC output.

3. Components Used

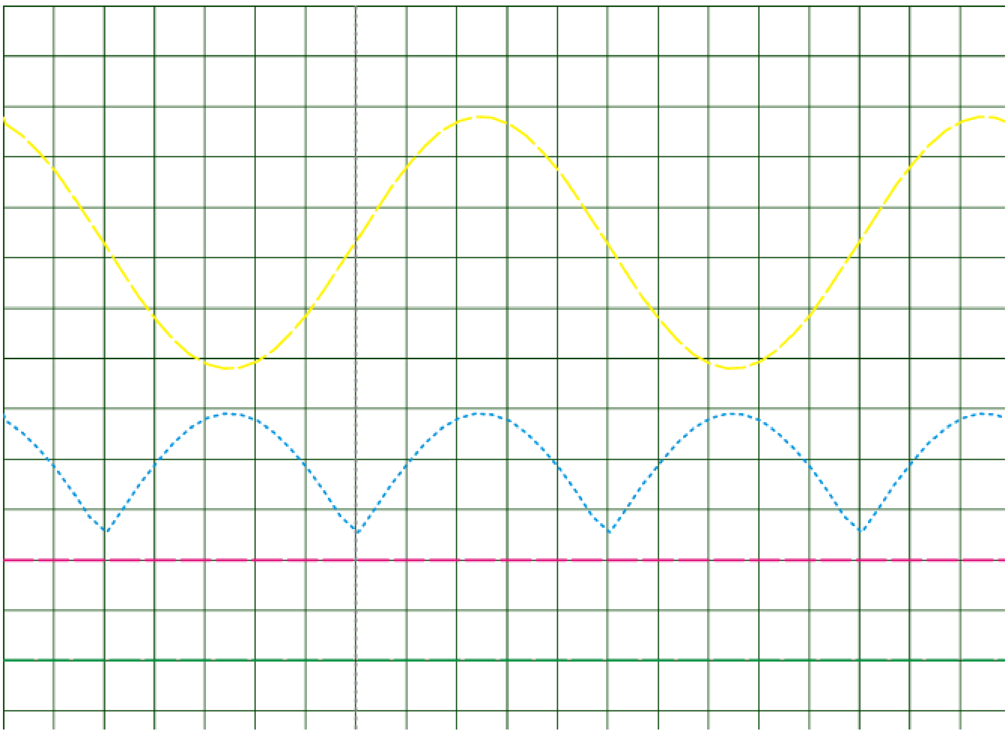
Sr. No.	Component	Specification	Quantity
1	Diode	1N4007	2
2	Voltage Regulator	7805	1
3	Electrolytic Capacitor	1000 µF, 25 V	1
4	Electrolytic Capacitor	100 µF, 25 V	1
5	LED	Red LED	1
6	Resistor	330 Ω	1
7	3-Pin Terminal Block	Input Connector	1
8	2-Pin Terminal Block	Output Connector	1

4. Circuit Description –

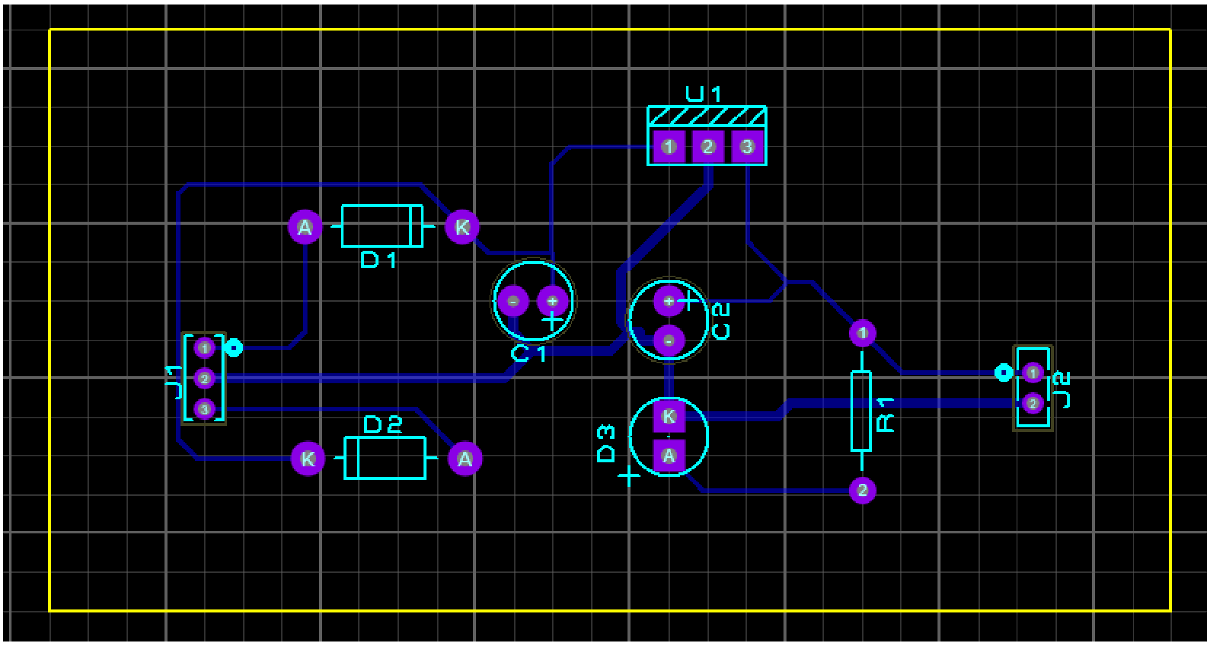
I) Schematic:



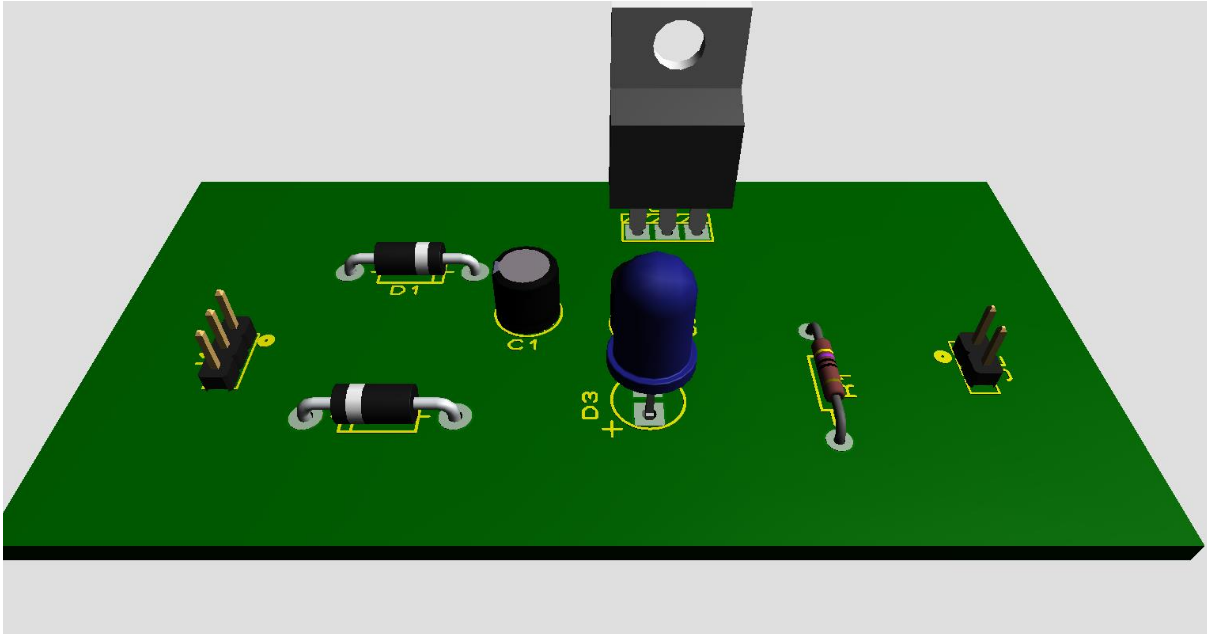
II) Waveform:



III) PCB Layout:



IV) 3D Visualization:



The AC input from a centre-tapped transformer is applied to the rectifier section. Two 1N4007 diodes perform full-wave rectification. A 1000 µF capacitor filters the pulsating DC and reduces ripple. The filtered voltage is supplied to the 7805 voltage regulator IC, which provides a regulated +5 V output. A 100 µF capacitor improves output stability, while an LED indicates power availability.

5. Working Principle

1. AC voltage is supplied through the 3-pin input connector.

2. Diodes D1 and D2 conduct alternately during positive half cycles.

3. The output becomes pulsating DC.

4. Capacitor C1 smooths the DC voltage.

5. The 7805 regulator converts the unregulated voltage into a constant +5 V DC.

6. Capacitor C2 further stabilizes the output.

7. The LED glows when the circuit is powered.

6. Input and Output Specifications

- Input Voltage: 12-0-12 V AC
- Rectifier Type: Full-Wave Centre -Tapped
- Output Voltage: +5 V DC
- Voltage Regulator: 7805
- Indicator: Red LED
- PCB Type: Single Layer Through-Hole PCB

7. Software Used - Proteus Design Suite

8. Hardware Fabrication Process

1. Designed the schematic in Proteus.

2. Generated PCB layout.

3. Printed PCB pattern on copper-clad board.

4. Performed PCB etching.

5. Drilled component holes.

6. Mounted components and completed soldering.

7. Tested the circuit successfully.

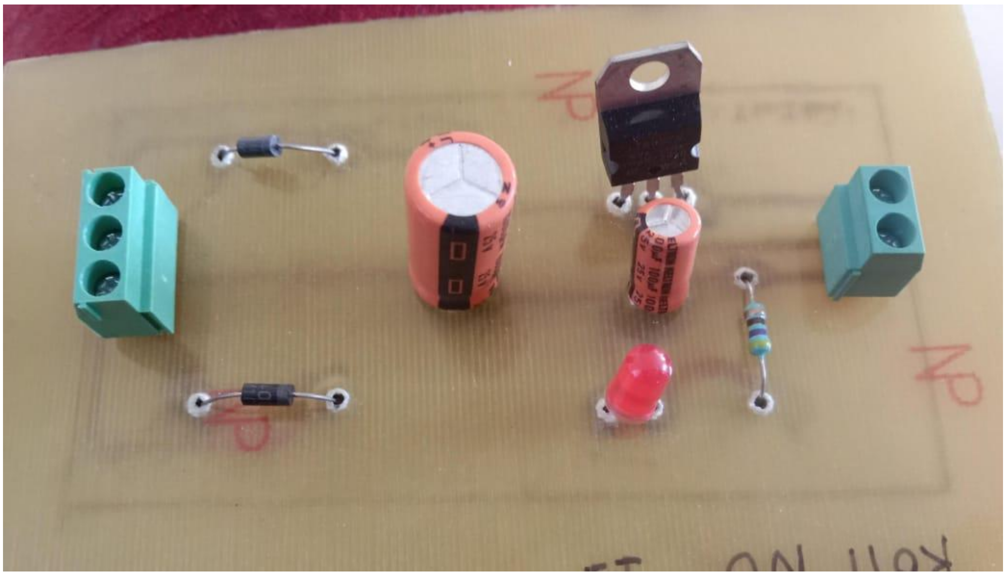


Figure – Top view of hardware

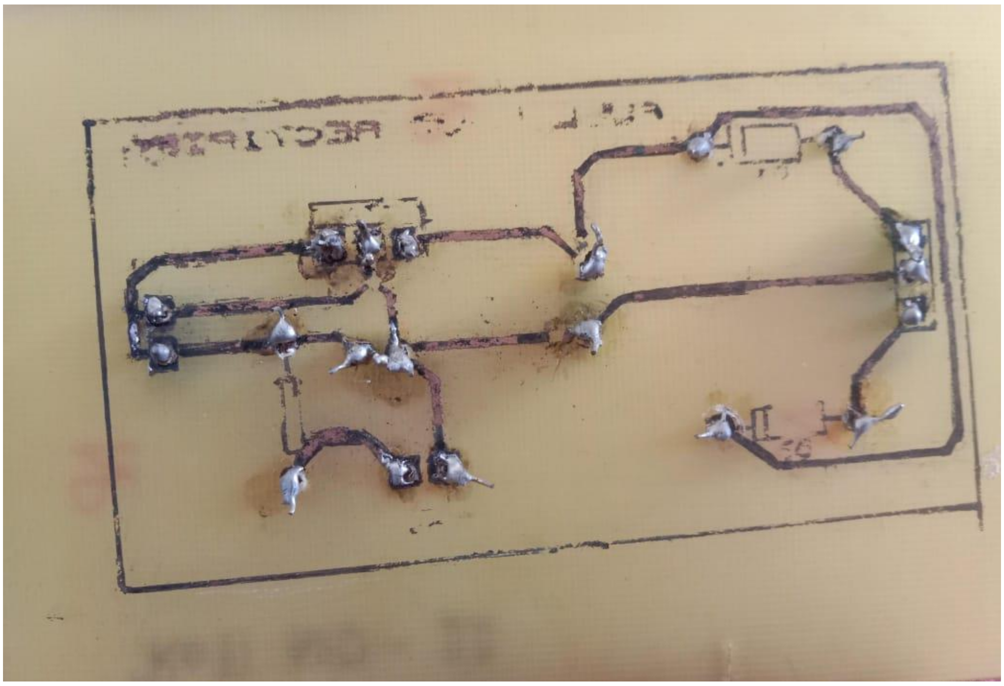


Figure – Bottom copper track view

9. Applications

- Power supply for microcontroller circuits.
- Embedded system projects.
- Educational laboratory experiments.
- Low-power electronic devices.

10. Conclusion

The Full-Wave Rectifier with 7805 Regulated Power Supply was successfully designed, fabricated, and tested. The circuit converts AC voltage into a stable +5 V DC output and demonstrates practical skills in schematic design, PCB design, fabrication, drilling, and soldering.